CHALLENGES IN OPEN BANKING – WHAT ARE THE PRACTICAL STEPS TO BE TAKEN NOW?

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The advance of open banking is an inevitable fact and it will inevitably change the digital landscape of the banking industry. However, open banking is not without its share of challenges with, for example, the Payment Services Directive (PSD2) and the General Data Protection Regulation (GDPR) in the EU testing the cybersecurity capabilities of banks,[1] *and the lack of enthusiasm by consumers to this 'game changing' concept slowing its progress and perceived success. As in the EU, in Australia the changes are driven by regulatory regimes, rather than the market, which sets the dominant agendas unlike in the US where they are market driven. The questions are how does each of the participants in open banking position itself within the new landscape and what can drive the further progress of open banking? This paper is taking an albeit brief look at the developments in the EU and Australia and looks at the issues of the past, present and future which need to be mastered in order to profit and function in the new regulatory environment. The lessons are that inaction - that is sitting on the fence - is dangerous and the time is now to plan and take action to occupy a prominent place in the open banking ecosystem before the arrival of the BigTechs and that acceptance by consumers is required in order for open banking to reach its full potential.*

I INTRODUCTION

Change in the reality of everyday life is not new, it has been a feature of human endeavours. However what changes do have in common is that the result is not foreseeable at the time when the change occurs. We are in the middle of a change namely brought about the advances of the IoT which has changed many aspects

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[1] Capgemini Research Institute & Efma, 2020, 'World Retail Banking Report' 18 <https://worldretailbankingreport.com/wp-content/uploads/sites/3/2020/06/World-Retail-Banking-Report-2020.pdf>.

of how we do business including - and importantly to this paper - the banking business. This paper endeavours to pause for a moment and take a reflective approach to where we stand and where the industry is possibly heading.

As change is always inevitable the past must not be forgotten as the most important aspect is to lean from past mistakes and take a proactive rather than reactive approach to change. Moving one step backwards we have seen globalisation which was hailed the saviour of trade by widening the reach from closed domestic economies to the international arena. It is seen decades later as only a temporary phenomenon. Ralston in relation to globalisation noted: "there are some remarkable successes. Some disturbing failures and a collection of what might best be called running sores."[2]

There is no secret that every change is also accompanied by uncertainty which generates opportunities but also problems. Not surprisingly, change always is followed by more change that is ideology and trade sometimes make strange bedfellows and what appears to be new becomes a habit. John Maynard Keynes after the First World War in 1919 commented: "The power to become habituated to his surroundings is a marked characteristic of mankind. Very few of us realize with conviction the intensely unusual, unstable, complicated, unreliable, temporary nature of the economic organisation by which Western Europe has lived for the last half century."[3]

Today it is technology that is the IoT which binds us all together. In essence change is thrust upon us by change in economic activities and perception of how the change ostensibly improves our ability to function in a new world or reality. But in essence as we see from the dark side of the IoT it weakens our ability to choose. It is also observable in times of great changes that "the belief that we do not have choices is a fantasy, an unfortunate indulgence in abdication."[4] As the IoT habituated society even banking is not immune and hence the traditional way to bank, that is face to face banking, has been challenged and is being overtaken by talking to faceless people. As an example, "Sometime, Australia's youngest bank – a neobank - is opening to the public, but there won't be a branch customer can visit, even if they want to. Instead, the unusually-named 86 400 will launch by making its app available to people with iPhones and Android

[2] John Ralston Saul, The Collapse of Globalism and the reinvention of the World, Viking Publishing, 2005, 3.
[3] John Maynard Keynes, The Economic Consequences of the Peace (London McMillan and Co (1919) 1.
[4] Saul (n 2) 13.

smartphones."[5] It sound eerily true what Saul noted in the demise of globalism when he observed:

> When a grand idea of ideology is fresh and the sailing is easy, even the most serious proponents make all-inclusive claims on its behalf. This grand view makes it easier for them to impose the specific changes they want.[6]

This is where the danger lays as seen in the effects of the Royal Commission into banking where mistakes in banking practices were uncovered. Hence specifically in Australia new ways need to be found to instil confidence in the banking system. The issue is that we need to learn from past mistakes and make sure that the observation made by Saul in creating a viable open banking system are avoided namely:

> The true believers and sophisticated managers go on riding their great wave ever more relentlessly and with remarkable but increasingly meaningless skill, while a growing percentage of the citizenry draw ever further back on the shore in a precautionary manner.[7]

The point is a financial system needs to be created which not only fulfills the needs of business that is the consumer but also allows banks to be competitive. Interestingly still in 2019 despite the creation of open banking in the EU there are still views that "banks still need retail locations with staffing even for the inline and cashless formats. Further, capitalizing on The Network Effect may mean actually increasing branch locations across a market."[8] The point is "we need new innovations focused on protections that are more conversation driven and transparent."[9] The question is are open-banking principles truly unique and wholly novel? In reality, the Organisation for Economic Co-operation and Development ("OECD") has sought to address these risks ex ante, through

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[5] Clancy Yates, August 17, 2019, Sydney Morning Herald, 'Meet the 'neobanks' trying to shatter the big four banking oligopoly' <https://www.smh.com.au/business/banking-and-finance/meet-the-neobanks-trying-to-shatter-the-big-four-banking-oligopoly-20190815-p52hiu.html>

[6] Saul (n 2) 5.

[7] Ibid., 13.

[8] Sean Keathley, November 16, 2019, 'Right-sizing branch networks to leverage perceived presence with the Network Effect', Adrenaline <http://www.adrenalineagency.com/blog/formats-efficiency/?financial-brand-2019-november-n>.

[9] Mary Meehan (Contributor), Forbes, 'Data Privacy Will Be The Most Important Issue In The Next Decade' <https://www.forbes.com/sites/marymeehan/2019/11/26/data-privacy-will-be-the-most-important-issue-in-the-next-decade/#25e418ed1882>.

guidelines on data protection, including constraints on the data that controllers and processors collect and use, what personal data they cannot collect.

The fact is that banks have lost the privileged position as financial institutions and changes are thrust upon the financial system to be more competitive. The general view in relation to the threats to banking is:

> The positioning of traditional banking models along the value chain is at higher risk now than ever before. PSD2 (Payment Services Directive) is disrupting banks' previously strong and integrated business, right down to the ownership of the client relationship and interface. Thus, we believe that the financial services value chain will significantly change, with a stronger distinction between different roles.[10]

The protection of the more traditional banking system is that banks still enjoy a higher trust by their customers in relation to data protection, but the question is how far will customer loyalty be a defence of the banks against other players entering the financial market. The issue is that big tech companies also have a huge customer base and could offer financial services without requiring a banking licence. A sign that the financial sector is rapidly changing is demonstrated by the fact that:" Both Visa and Mastercard Inc. have been seeking to move beyond card payments in recent years to fuel their rapid revenue growth. Mastercard bought a payments platform owned by Nets for $3.2 billion last year, using its biggest-ever acquisition to move further into so-called account-to-account payments."[11]

The EU introduced the PSD2 system over two years ago and the Australian government has also passed the Consumer Data Rights legislation (CDR)[12] which according to the government will allow consumers to take charge of their data. The government's stated policy rationale for the CDR is:

> Through requiring service providers to give customers open access to data on their product terms and conditions, transactions and usage, coupled with the ability to direct that their data be shared with other service providers, [the government

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[10] Roland Berger, 2019, Focus, 'Adapt or die? Why PSD2 has so far failed to unlock the potential of Open Banking', 11 <https://www.rolandberger.com/publications/publication_pdf/roland_berger_psd2.pdf>.

[11] American Banker, 'Visa to buy plaid for $5.3 billion in bid to reach start-ups' <https://www.americanbanker.com/articles/visa-to-buy-plaid-for-53-billion-in-bid-to-reach-startups>.

[12] Treasury Laws Amendment (Consumer Data Right) Bill) 2019 <https://www.legislation.gov.au/Details/C2019B00025>.

> expects] to see better tailoring of services to customers and greater mobility of customers as they find products more suited to their needs.[13]

The aim is to roll the CDR out "sector by sector across the economy, beginning with the banking sector" (or open banking).[14] However, its implementation has been delayed due to the "Australian Competition and Consumer Commission pushed back the start date for the government's open banking reforms from February to July and now most likely to the end of the year,"[15] with data of consumers concerning mortgages and personal loans being available to be shared from 1 November 2020.[16] The reason for the delay is that the ACCC was not convinced that the system was resilient enough against cyber-attacks. The ACCC noted:

> We at the ACCC have formed the view that because of these potential issues, and because we want to take time to work through them properly, that it's better to push out the start date to allow the whole ecosystem to really have undergone that … more extended security testing.[17]

Arguably the ACCC appears to be at ease with the reforms of the banking system as long as security is guaranteed. The problem is not how the banks safeguard data, but the problem lies in the pipes which transport data form the banks to fintechs that is Trusted Third Parties (TTPs).

One of the problems which the Australian economy - and the government needs to work it out - is whether the desire to open the Australian banking sector to increase overseas competitors is an advantage to the Australian economy and the viability of the four big banks is not jeopardised as they stand at the moment. "A new cohort of global fintech players are eyeing the Australian market, with UK open banking software player TrueLayer planning to launch here in 2020, opening the door for its European clients to access Australian consumers."[18]

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[13] Exposure Draft Explanatory Materials, Treasury Laws Amendment (Consumer Data Right) Bill 2018 (Cth), 3 para 1.3.

[14] The National Law Review, June 05, 2020, 'Clarity for Consumer Data Right Participants on Regulatory Obligations: Is Your Business Ready?' <https://www.natlawreview.com/article/clarity-consumer-data-right-participants-regulatory-obligations-your-business-ready>.

[15] Clancy Yates, December 22, 2019, The Sydney Morning Herald, 'Open banking reforms delayed as security tests drag on' <**https://www.smh.com.au/business/banking-and-finance/open-banking-reforms-delayed-as-security-tests-drag-on-20191220-p53lxv.html**>.

[16] The National Law Review (n 14).

[17] Yates (n 15).

[18] Emma Koehn, October 14, 2019, The Sydney Morning Herald, 'Global fintechs set sights on Australia as open banking nears' <**https://www.smh.com.au/business/small-business/global-fintechs-set-sights-on-australia-as-open-banking-nears-20191011-p52zpe.html**>.

Of course, an argument is that the Royal Commission has uncovered practices which are not sound and hence unethical which competitors would arguably they can alleviate. The bigger picture is how new competition will not only affect the banks but the share market and hence superannuation portfolios. This will be a crucial issue as bank shares which are a big proportion in any portfolio[19] might drop in conjunction with the profits of banks in Australia affecting "the man in the street" hence indirectly the whole economy.

A further and very important observation is that "the E.U. regulatory regimes, rather than product innovation, set the dominant agendas. Simply complying with regulations and establishing new stakeholder categories have been the dominant concerns of banks and other stakeholders."[20] As a result, new stakeholders emerged as well as new platforms such as the platform as -a-service business model (PaaS). The effect of the new models will have a deep effect on the financial sector bringing new challenges to the fore.

The major problem is that also in the EU delays have made full implementation impossible. The issue is that strong customer authentication (SCA) has not been implemented by many merchants and hence compliance is an ongoing issue. In addition TTP's asked for an extension of time as the APIs of banks are not ready and hence putting the TTP's at risk.[21]

In sum it is argued that in times of change sound business practices need to be observed whether it is in the manufacturing or banking industry. Once sound practices are lost it is hard to refocus as it takes time and the changes need to be not only a top down effort but a combined one focusing on all levels of the business. The difficulty to change has been demonstrated that even after the Royal Commissions findings, transgression are still occurring and at best can be described as symptoms of change and at worst the inability to change a business culture. The issue is how will open banking avoid the transgressions which have been laid bare by the Royal Commission that is have banks learned the lessons?

In general, Open banking requires a tripartite approach in the new ecosystem between banks, third party service providers (TPPs) and regulators, with the consumer (the end user) central to all of these. Banks and TPPs have to put in place a method which takes into account any relevant regulatory requirements.

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[19]Commonwealth Bank, Westpac, National Australia Bank and ANZ Bank have a combined market capitalisation of about $380 billion, a fifth of the ASX 200.

[20]Mercator Advisory Group, December, 2019, 'The Emergence of API Platforms: Open Banking and Payments Drive New Business Models' 4- < https://www.mercatoradvisorygroup.com/Reports/The-Emergence-of-API-Platforms--Open-Banking-and-Payments-Drive-New-Business-Models/>.

[21]Berger (n 10) 6.

In turn these regulations must be clear. The sharing of data by the consumer with TPPs is the cornerstone of the open banking system upon which everything else depends. The system must therefore confirm to consumers that open banking will deliver the required security to safeguard their personal data (be safe) produce the beneficial services that they seek, (of benefit) and will be delivered faster and preferably cheaper than the old method of banking. In other words, consumers must be convinced that open banking is a step up from the current system, which in part is or was based on face to face conversations. In order to be able to deliver their services to consumers, TPPs must be provided the requisite consent by consumers and then access by the banks, who hold the relevant data.

In order to provide access to TPPs, banks need to verify that their system is protected in relation to the inclusion of TTP's and a compatibility between the users is established. The issue is that all systems need to be able to work together in order to supply the desired outcome. Importantly:

> Building end user trust this new ecosystem will require cross-industry co-ordination and potentially investment, not just in education and awareness campaigns around how to enjoy the benefits of new TPP services, but also through initiatives to provide reassurance around the TPP discovery and selection process.[22]

In addition, conversation around the shape of the products which will be on offer needs to take place. For some in the industry it merely means to improve the existing processes. The second consideration is to take banking services outside the traditional banking channels such as payments into the marketplace of consumers.[23] However the focus must be on the creation of an API which is user friendly, adaptable and secure. A special challenge is how to operate in the transnational landscape where APIs are not compatible. As it stands now "The United Kingdom set up the Open Banking Implementation Entity (OBIE), while the Berlin Group launched its NextGenPSD2 standard, France developed STET, whilst Poland has developed the Polish API. While most European banks now adhere to the NextGenPSD2 standard, there is room for individualization. For example, the NextGenPSD2 standard alone has six possible variants of an authentication procedure."[24]

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[22] Mastercard, White Paper, 2019, 'Delivering on the promises of Open Banking' <https://www.mastercard.us/content/dam/mccom/en-us/documents/white-paper-delivering-promises-open-banking.pdf> 12.
[23] Ibid., 17.
[24] Berger (10) 7.

Hence this article so far has attempted to point out the elements which are crucial in the open banking system. In the end the question consumers will ask is, what's in it for me? Will open banking be superior to traditional mechanisms and will it satisfy consumer needs and not merely serve the internal purpose of increasing the bottom line?

What is clear is that there is no doubt that open banking will fundamentally change the way we interact in the financial sector. It will bring new opportunities but also new challenges to the whole ecosystem. Importantly it is serving as the pilot for open finance, which will seek to expand the focus of financial data sharing to other areas such as mortgages, investments and insurance.[25]

Open banking in essence caters for five different players namely large banks national or international, smaller regional banks, monoliners specialising in a specific category of products, and bigtech firms.[26] The benefits for banks - traditional and smaller ones - will have the opportunity under the new ecosystem to understand their clients better as they can view the accounts of their clients which they keep in different banks. Hence the opportunity is there to give better advice as the banks know the full financial ability of the consumer. The biggest advantage if realised is:

> Instead of relying on e-commerce payment methods or the large card networks at the point of sale (POS), who take a cut, banks could offer account-based e-commerce and POS payment solutions via PSD2 APIs. Doing so could enable these players to strengthen their position as the relationship expert at the client interface.[27]

Monoliners have traditionally lacked a broad financial view of their clients. Open banking allows the monoliners to keep track of account balances hence being product experts they can assess their risks better. Neobanks – a new invention – need to focus in growing a client base but they do not have the ability to offer a full range of financial products such as mortgages. Bigtechs on the other hand already have an established brand hence if companies like Facebook will offer financial services, they can rely on their huge client base[28].

To that end in the first part the requirements of privacy laws will be discussed specifically in light of the EU legislation which appears to be the model to be

[25] Financial Conduct Authority, December 2019, 'Call for Input: open finance' <https://www.fca.org.uk/publication/call-for-input/call-for-input-open-finance.pdf>.
[26] Berger (n 10) 19.
[27] Ibid.
[28] Ibid., 21.

followed only in light of obligations by banks.[29]. The second part will have a brief look at the UK and European experience. The third part will discuss the issues of the creation of trust in the open banking model. The fourth part will discuss issues in relation to the emergence of Application Programming Interface (API) platforms, followed by a conclusion.

II PRIVACY REQUIREMENTS

Open banking has been described by the Basel Committee on Banking Supervision as "relat[ing] to customer-permissioned data sharing where the customer initially grants permission to a third party firm ("third party") to access their data, either directly, or through the customer's bank."[30] In addition the Committee noted:

> This sharing of customer-permissioned data by banks with third parties could be leveraged to build applications and services that provide faster and easier payments, greater financial transparency options for account holders, new and improved account services, and marketing and cross-selling opportunities.[31]

The committee also examined 17 jurisdictions specifically focusing on "supervised banks and customer-permissioned data."[32] An important point is that the data must be customer-permissioned but arguably must be in compliance with data protection laws.

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[29] For simplicity's sake "banks" has been used in this paper instead of account servicing payment service providers ("ASPSPs"), being any payment service provider providing and maintaining a payment account for a payer. This includes banks and building societies. See:
Directive (EU) 2015/2366 of the European Parliament and of the Council of 25 November 2015 on payment services in the internal market, amending Directives 2002/65/EC, 2009/110/EC and 2013/36/EU and Regulation (EU) No 1093/2010, and repealing Directive 2007/64/EC (Text with EEA relevance) (PSD2).
<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32015L2366>.
Meanwhile the OBIE has defined ASPSPs as follows: "Account Servicing Payment Service Providers provide and maintain a payment account for a payer as defined by the PSRs and, in the context of the Open Banking Ecosystem are entities that publish Read/Write APIs to permit, with customer consent, payments initiated by third party providers and/or make their customers' account transaction data available to third party providers via their API end points; Therefore requiring an online activity
<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32015L2366>.

[30] Basel Committee on Banking Supervision, November 2019, 'Report on open banking and application programming interfaces' 4- <www.bis.org>.

[31] Ibid., 8.

[32] Ibid.

Open banking will no doubt create data. As a matter of fact "Banks sit on an invaluable treasure trove of personal data." [33] It is not a secret that data has value as shown by the data miners such as Facebook and arguably in the future some financial institutions might be tempted to "cash in" on the data they hold. However, data protection laws regulate the use of personal data such as purchasing habits, travel options and special category data such as trade union memberships or health and medical transactions. Art 4(1) of the GDPR defines personal data as any information relating to an identified or identifiable natural person ('data subject'). The definition of personal data implies that there is a clearly distinguishable category of non-personal data. As personal data is used every bank (and other financial institutions affected by open banking) need to have in place a lawful basis in order to process and hence share data. This is specified in the GDPR as "legitimate interest" in article 5(b). In order to rely on legitimate interests as a lawful basis, organisations must carry out a "balancing test" or legitimate interests assessment to consider whether its use of the data is appropriate and proportionate to the impact that it will have on the individual whose data it is. In addition, "even if it is considered that it is appropriate, you would need to offer an opportunity to object."[34] It is now established in the EU that "merely serving purely commercial interests, including following the (buying) behaviour of (potential) customers, would not qualify as a legitimate interest in any event,."[35]

Additionally, any bank would need to be aware of the potential of data contamination by TTPs whether accidentally or by design. Hence the chosen API (specifically if blockchain is chosen) needs to take the verification and unchangeability of data into consideration. In addition, TTPs might transfer data onto their own systems and hence could be open to data theft and therefore relevant safeguards need to be in place. Arguably banks could consider asking TTPs for bank guarantees in order to be able to cover any potential claims by customers for misuse of data.

In Europe, one of the questions that has continually been asked since the implementation of GDPR was, how will it interact with PSD2, given that they are not exactly aligned? In July 2020 the European Data Protection Board ("EDPB")

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[33] Berger (n 10) 9.
[34] Eversheds Sutherland, December 2019, 'Wrapping up for the year, 12 days of payments' <https://www.eversheds-sutherland.com/global/en/what/articles/index.page?ArticleID=en/Commercial/12-days-payments>.
[35] Ibid.

published draft guidelines on the interplay of GDPR and PSD2 for public consultation, with a particular focus on the processing of personal data by AISPs and PISPs.[36] The topics covered include, lawful grounds for further processing under PSD2, explicit consent under PSD2 v consent under GDPR, the processing of silent party data, special categories of personal data under PSD2 by AISPs and PISPs and Data minimisation.[37] Given the sometimes narrow positions adopted by the EDPB (e.g. the proposed position concerning Article 6 1(b) in relation to the processing of personal account data for payment initiation and account information services) it will be very interesting to see how parties involved in open banking will respond.

As in Australia the open banking system will be rolled out by the end of the year, it is instructive to look albeit briefly at the UK and general EU experience

III The UK And General Eu Experience

Traditionally in the UK, the older and larger retail banks have been typically cautious when it comes to changing their ways. After all, if it isn't broke, why fix it? However, in 2016 the Competition Market Authority ('CMA') published the final report of its market investigation into retail banking in the UK[38] that it had started in 2014 in which it reasoned that such banks would only improve their services, reduce their pricing or invest in new products if they feared a loss of customers or saw an opportunity to win new ones.[39] To tackle this issue the CMA outlined a comprehensive package of remedies, one of which was the use of an open API banking standard to share information such as the location of branches and terms for banking products in a secure environment[40] and also to share a customer's transaction history with their informed consent to enable potentially beneficial services for them.[41] In parallel to the CMA's final report, PSD2 entered

[36] European Data Protection Board, July 22 2020, 'Guidelines 06/2020 on the interplay of the Second Payment Services Directive and the GDPR – version for public consultation' <https://edpb.europa.eu/our-work-tools/public-consultations-art-704/2020/guidelines-062020-interplay-second-payment-services_en>.

[37] Ibid.

[38] Competition & Markets Authority (CMA), August 9 2016, 'Retail banking market investigation - Final report' <https://assets.publishing.service.gov.uk/media/57ac9667e5274a0f6c00007a/retail-banking-market-investigation-full-final-report.pdf>.

[39] CMA, August 9 2016, 'Making banks work harder for you' <https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/544942/overview-of-the-banking-retail-market.pdf>.

[40] CMA (n 38) paras 166-167.

[41] Ibid.

into force in January 2016 and applied from 13 January 2018, with some matters such as strong customer authentication applying from a later, to be determined, date.[42] Amongst the changes ushered in by PSD2 was the regulation of two new types of providers for the first time, providers of account information services (AISPs) and payment initiation services (PISPs) (together TPPs).[43]

In 2017 the CMA issued the Retail Banking Market Investigation Order (the "Order") which included a mandate that the top 9 UK banks form an Implementation Entity to set up and maintain open API, data and security standards, governance arrangements and certain customer redress mechanisms in the UK (the "OBIE').[44] The work of the OBIE concerning TPPs and ASPSPs in the UK continues.

It needs to be noted that TPPs were already in existence prior to PSD2. What this new regime serves to do is to fully open the door to a TPP accessing the data of a banks client and hence create a new type of requirement of, and also a benefit to, a TTP which can now transfer its services into the open banking protocol and become a valuable part of the ecosystem. Further, it serves to seek to shift the 'ownership' of data to the consumer from the bank"[t]he big shift under PSD2, however, is the setting of a regulatory framework to transfer ownership of client data from banks to clients, requiring all banks to implement APIs. Financial data has become a commodity, and clients are free to choose which provider can use it."[45]

It is certainly feasible to believe that the new legal framework will create a foundation upon which regulated parties can eventually deliver truly game changing services and options to consumers and businesses, regardless of location. The question is when this will realistically occur? Eversheds Sutherland

[42] European Banking Authority (EBA), June 21 2019, 'Opinion of the European Banking Authority on the elements of strong customer authentication under PSD2, para 7- <https://eba.europa.eu/sites/default/documents/files/documents/10180/2622242/4bf4e536-69a5-44a5-a685-de42e292ef78/EBA%20Opinion%20on%20SCA%20elements%20under%20PSD2%20.pdf>.

[43] EBA, June 4 2020, 'Opinion of the European Banking Authority on obstacles under Article 32(3) of the RTS on SCA and CSC, para 1 - <https://eba.europa.eu/sites/default/documents/files/document_library/Publications/Opinions/2020/884569/EBA%20Opinion%20on%20obstacles%20under%20Art.%2032%283%29%20RTS%20on%20SCA%26CSC.pdf>; also see, FCA, 'New regulated payment services Account Information Services (AIS) and Payment Initiation Services (PIS)'<https://www.fca.org.uk/firms/new-regulated-payment-services-ais-pis>.

[44] CMA, 2017, 'RETAIL BANKING MARKET INVESTIGATION; The Retail Banking Market Investigation Order 2017' part 2 - <https://assets.publishing.service.gov.uk/media/5893063bed915d06e1000000/retail-banking-market-investigation-order-2017.pdf>.

[45] Berger (n 10) 8.

International described the current state of open banking in the UK as consisting of three ghosts; the one past, present and yet to come.[46] Whilst PSD2 and the OBIE may have brought the concept of open banking in the UK to life, after a number of years in the making 260 regulated providers have registered with the OBIE of which 186 are TPPs with the rest being account providers.[47] Considering that the UK is a big economy the number of registrations does not appear to be an overwhelming response. Currently it can be argued that banks have not risen to the challenge as relatively few have positioned themselves as a TTP,[48] and their promotion of open banking in the UK has been somewhat reserved. However, given that the Order only related to the top 9 UK banks and that it did not mandate the actual promotion of open banking by them or the OBIE, let alone banks in general, query whether these parties actually need to do any more than what they are currently doing. Given the circumstances it is reasonable to conclude that there is currently more to gain by being a TPP in the bank/TPP relationship and therefore it will be the TPPs (and TSPs) not the banks that we will see actively promoting the benefits of open banking to consumers and SMEs in the UK in the short term. This approach may change when banks move away from being 'just a bank', e.g. if banks overwhelmingly decided to also become TPPs, this would be one way which would result in them having real skin in the game. However, only time will tell.

It also appears that banks across Europe are still working on their account access solutions, which will result in TPPs being able to deliver their services in an uninterrupted, safe, effective and convenient fashion, something that was intended by PSD2. In June 2020 the EBA published an opinion on obstacles to the provision of TPP services under PSD2[49] and sought to provide clarification on matters such as mandatory redirection at the point of sale and multiple SCAs. Of note is the EBA's expectation that the NCAs will take the clarifications contained in the opinion into account when monitoring the compliance of the interfaces provided by the banks with PSD2.[50]

The road to successful open banking in Europe is obviously a long one. However it is equally obvious that the APIs must conform to the same

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[46]Eversheds (n 34).
[47]OBIE, 'Highlights June 2020' <https://www.openbanking.org.uk/wp-content/uploads/JUNE-Monthly-highlights.pdf>.
[48]Berger (n 10) 9.
[49]EBA (n 43).
[50]Ibid.

requirements, such as the ones outlined by the OBIE and the Berlin Group, if we are to have any hope of ultimately achieving a uniform level of consistency by which European open banking will be allowed to prosper. Initial segmentation is fine, but a uniform API standard within a realistic timeframe needs to be the target. If this occurs a global standard is not out of the question and by being first to align an agreed European standard could be the framework for a global one. As noted by Berger, maybe it will only be a matter of time before all proposed standards will eventually converge.[51] In addition, discussions with the relevant participating industry bodies cannot be forgotten. Their input and feedback on lessons learned from their journey is vital. Further, the regulations which apply to the open banking movement are numerous and complicated and not in all instances easy to understand. Due to this, various EBA guidelines and regulatory technical standards experienced delays and several European countries (e.g. Belgium, Netherlands, and Spain) failed to meet the transposition deadline of 13 January 2018.[52] An example of an issue which has caused particular uncertainty in the UK relates to offering direct eIDAS identification."[53] In addition, screen-scraping and 'reverse engineering' is a problem that the RTS seeks to address with it not being permitted beyond 14 September 2019 and banks thereafter being permitted to refuse access that way.[54] However in October 2019 the EBA recommended that this timeline has been extended for e-commerce card-based payment transactions to 31 December 2020,[55] whilst in the UK recently extending it to 14 September 2021.[56]

The Ghost of Open Banking yet to come is how banks will deal with the new entrants into the market whilst existing TTPs will attempt to expand their business hence issues of "cutting into the business" will occupy banks in the future. Therefore, it is obvious that the open banking system is and will remain fluid presenting new challenges in the future. Do banks decide to just do what

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[51] Berger (n 10) 7.

[52] Innopay, February 21 2019, 'PSD2 licensing: solving the puzzle of becoming a Third Party Provider' <https://www.innopay.com/en/publications/psd2-becoming-a-third-party-provider>.

[53] Eversheds (n 34).

[54] EBA, 'Final Report: Draft Regulatory Technical Standards on Strong Customer Authentication and common and secure communication under Article 98 of Directive 2015/2366 (PSD2)' 118 - <https://eba.europa.eu/sites/default/documents/files/documents/10180/1761863/314bd4d5-ccad-47f8-bb11-84933e863944/Final%20draft%20RTS%20on%20SCA%20and%20CSC%20under%20PSD2%20%28EBA-RTS-2017-02%29.pdf>

[55] EBA (n 43).

[56] FCA, (last updated) June 18 2020, 'Strong Customer Authentication' <https://www.fca.org.uk/firms/strong-customer-authentication>.

they are obligated to do, or do they decide to embrace the change and partner up with TPPs when business efficacy requires? What will their market driver be? Much would no doubt have to do with customer expectations and confidence in the system, plus the commercial benefits that could be achieved. After all the banks need to have a viable business model which not only attracts customers but is also profitable for the banks and importantly keeps the bank in a competitive framework. As an example will open APIs form part of this business model?[57] In addition, as noted in Part 1 above, privacy laws will dictate to an extent how the model will work as customers need to consent to having their data made available on an API in addition to the API owner protecting the entrusted data. Another point to consider of course is supply/demand. If customers are not asking for it, banks and TPPs will be slower to invest in new products and services. However, in order to get consumers excited about open banking they need to know that it is secure and be convinced that it benefits them. Based on what is happening in Europe, it appears that consumer 'demand' needs more investment by those on the supplier side.

"Open banking is the first data sharing and data access initiative of this scale to be underpinned by legislation requiring industry institutions to provide access to other providers."[58] Whatever the view on its success to date, the future of open banking in Europe is bright with notable activity from technical companies[59] and it appears that it is the start of something greater with open finance, which expands upon the data sharing and third party model seen in open banking to other areas such as insurance, savings, mortgages, investment and consumer credit [60] appearing on the horizon in the UK.

[57] Whilst Open APIs are different to Open Banking, they form its foundation.

[58] FCA (n 25) para 2.12.

[59] In its response to the EC's consultation on the digital finance strategy/action plan in June 2020 the EBA noted "that technology companies have become more active in the payments market than in the banking sector, potentially leveraging the PSD2 and data sharing through application programming interfaces (APIs) to enter the payments market given their existing customer base, scaling experience and available technology tools", EBA, June 26 2020, 'EBA response: EC consultation on the digital finance strategy/action plan' response to EC consultation on the digital finance strategy/action plan' <https://eba.europa.eu/sites/default/documents/files/document_library/About%20Us/Missions%20and%20tasks/Correspondence%20with%20EU%20institutions/2020/886668/EBA%20Response%20to%20EC%20DFS%20consultation%20260620.pdf>.

[60] On 17 December 2019 the FCA issued a Call for Input on the opportunities presented by open finance with input accepted until 01 October 2020; FCA (n 25).

IV The Re-emergency Of Api Platforms

There is no doubt that in a regulatory environment technical complexity and to an extent an overregulation is an inevitable hurdle which needs to be overcome in order to exploit new and innovative revenue opportunities for the finance sector. One issue however is that sight of the ones who are the subject of revenue creation namely the consumer cannot be overlooked. If consumers reject the open banking system, open APIs will be of little value insofar as this method of data sharing for this particular purpose is concerned. Simply put the consumer or customers will need to see a value in open banking compared to the current model. It has been argued that:

> it is worth repeating that although both the European Union's revised Payment Services Directive (PSD2) and the United Kingdom's Competition and Markets Authority (CMA) Open Banking mandates are regulatory interventions intended to bring about market reforms, they are clearly not well aligned with the current business and revenue models associated with banking.[61]

The issue is that banks will no longer compete on the basis of their own products and services. With TTPs' involvement interaction between the consumer and the finance sector will no doubt be one stage removed.[62] What the push by regulators has achieved is that there are now new entrants in the payments sector. In addition to new banks, a variety of financial technology companies (fintech's) have been allowed to enter the market which due to increased competition will increase expenses for banks in the short term at least. From a consumer's point of view the open banking system will no doubt increase account holders' access and control to data and arguably it creates the potential of data portability and easier account switching. The question is whether consumers understand the benefits of account switching and whether data portability is an advantage to the less aware consumers. In essence account switching is certainly not a novel issue. Mortgage brokers have occupied this space and no doubt will continue to do so.

So far, the only certainty is that the regulatory and technical complexity mandated by regulators will have an effect on the financial sectors bottom line, certainly in the short term. In Australia the regulators and government have pushed the deadline out and again out and it is now assumed that by the end of

[61] Mercator (n 20) 5.
[62] Mastercard (n 22) 7.

2020 the regulatory framework will be in place. The main point is that the "dominance of the E.U. regulatory agenda has focused the banking industry on APIs as a necessary implementation strategy rather than a business strategy in its own right."[63] The obvious effect is that it has spawned new enterprises attempting to fill the gap in the provision of platforms and connections to expanding the launch of new tech providers but also well-established providers such as MasterCard and Visa are in the mix to look to provide a service to banks in the new "platformization" business model opportunity.[64]

APIs have in effect two functions namely an IT and a business strategy function. API's have been defined in many ways, but a useful definition is:

> An application programming interface (API) allows two different software solutions, operating both within and between businesses, to communicate together cooperatively. In financial transactions, the cooperation may be simple, such as allowing one application to receive data from the other to validate an account holder or a balance. The APIs can also perform complex functions, enabling applications to communicate back and forth to execute a complex business process as would be necessary to originate a loan or open a bank account.[65]

APIs have become the "de facto standard for sharing data and have become the "mechanism of choice for use over the internet to enable organisations that hold large amounts of data to become platforms for third party innovation."[66]It can be argued that the ability of an open API does not generate new paths or novel ways to access funds. What is novel is the fact that instead of being mostly paper based and information is stored in several Apps, all the information can now be combined into one interface hence saving time and being able to access information quicker and on a wider scale. In effect the API is the central pillar and banks are now more limited in being able to decide which third party is allowed access it, or must communicate with the bank on a different interface. Importantly, banks can also be TTPs should they choose that model.

Hence an open API model enables more efficient dealings between banks, tech companies and consumers. The challenge in the development of API interfaces is that customer and bank are still connected, and that customer satisfaction is

[63] Mercator (n 20) 5.
[64]Ibid.
[65]Ibid., 6.
[66]Financial Stability Board (FSB), February 14 2019, 'FinTech and market structure in financial services: Market developments and potential financial stability implications' 6 <https://www.fsb.org/wp-content/uploads/P140219.pdf>.

not sacrificed to the" god" of the internet. It is the balance between profitability and customer satisfaction which must be kept in mind. It would not be an exaggeration that everybody has had an experience with a bad interface or call centre which caused the prospective client or customer to look elsewhere. Open banking is no different and must rise to the challenge or face a reduced customer base.

A further issue is that corporate clients have a different need than retail customers and hence need to operate in a different interface as arguably the two are not compatible. This is so as the information corporate clients require would be of no value to retail clients and consumers. As an example, many US firms such as Wells Fargo have developed API's suiting their business model; "Wells Fargo has built a robust set of APIs ranging from account aggregation to account balance, account statements, image retrieval, making branch appointments, tax information, transaction-level detail, and access to Wells Fargo locations."[67]

However, the tools which a company might wish to use are subject to domestic legislation. As an example, Visa is using biometrics. In Australia the Commonwealth bank is looking at face recognition however that type of data is strictly owned and controlled by the government hence flexibility in the use of the API is required. Of real innovative nature is the supportive capabilities which open banking will be able to perform. Both MasterCard and Visa have taken advantage or will be taking advantage of open banking. "The recent offerings by Mastercard and Visa are notable for their broad scope, especially security, authentication, and anomaly detection capabilities, which safeguard the core solution functions."[68]

It is obvious that banks will still control and mange customers money. This will not change, certainly not in the short term. What is new is that the customer can allow third parties to access deposits and bring their experience into play. A good example is MasterCard and Visa who have for a long period approached banks who in turn issued credit and debit cards with a security number but managed by MasterCard or Visa. What has changed is that say MasterCard can now with the authority of the customer access banking details. The crux of the matter is obviously security that is the API system must protect the customer under the relevant privacy regulations such as the GDPR in the EU or the Australian privacy laws. This issue is not yet finalized. As an example, the EU PSD2 mandates that strong customer authentication processes (SCA) are in place

[67] Mercator (n 20) 8.
[68] Ibid., 11.

in order to guarantee secure communications. "The SCA regulations require two-factor authentication (2FA) when accounts are accessed online and when an account holder initiates an electronic payment transaction or takes any online action that might create the risk of payment fraud."[69]

If the open APIs are faulty, the whole system is likely to stall and/or collapse and hence this critical issue needs to be laid to rest. In addition, banks must have parity with the channels linking into their specific APIs as only then will the APIs perform properly and of course satisfy all concerned.

V Trust In Open Banking

Banks are by their nature trusted by consumers when it comes to financial services. For example, in Poland of those that took part in a 2019 survey by KPMG, 76% trusted banks when it comes to financial services, followed by Google with 27%.[70] However, trust in open banking is another matter. It must be remembered that it is not only trusting the system but getting to know the system in order to form an opinion that is trustworthy and know that it works. As of mid-2019, some 2 years after the introduction of the OBIE only one third of consumers had heard of open banking in the UK.[71] This was marginally up from one in four people as of 2017"[72]and even less understand the ramifications of opening up the financial sector. In Poland 61% of the consumers surveyed would not share their transaction history with other entities even if it meant obtaining a better financial offer.[73]

The trust and knowledge of open banking is especially important for Small and Medium sized businesses (SME's) who's business model relies heavily on intellectual property. The real change and hence the problems lie in the fact that the "issue of cybersecurity has transformed from a traditionally 'vertical' responsibility on IT departments to a 'horizontal' responsibility shared across the

[69] Mercator (n 20) 13.
[70] KPMG, March, 2019, 'PSD2 and Open Banking; Revolution or evolution?' 13 <https://assets.kpmg/content/dam/kpmg/pl/pdf/2019/04/pl-Raport-PSD2-i-Open-Banking-ENG.pdf>.
[71] Douglas Blakey, Retail Banker International, May 16 2019, 'UK open banking consumer awareness remains low' <https://www.verdict.co.uk/retail-banker-international/news/uk-open-banking-consumer-awareness-low/>.
[72] Chiara Cavaglieri, Which, October 29 2017, 'Open banking: 92% of the public in the dark' <https://www.which.co.uk/news/2017/10/open-banking-92-of-the-public-in-the-dark/>.
[73] KPMG (n 70).

supply chain end-to-end."[74] The interesting statistic is that "Consumers in Europe and Singapore — the other markets studied — are somewhat more open about sharing their data, but still, a majority of consumers (53% in Europe, 57% in Singapore) say they are unlikely to allow their bank to share their data."[75] It is interesting to speculate whether banks rely on these findings in order to wait and see before they need to act and devise a new business model.

A further issue which needs to be kept in mind is the issue of liability for data breaches. The EU has resolved that in the end "under PSD2, regardless of whether an unauthorised transaction has occurred as a result of [third party providers] TPP access, the rule of thumb is that a [Financial Institution] FI must reimburse the customer."[76] It is argued that this will always be the case as a customer in reality has no ability to work out where in the data chain the breach occurred and hence his remedy is to sue the FI directly. The FI is then obliged to discover the breach and if the contractual arrangements allow it can attempt to sue the relevant TPP.

The problem is that the API has to be robust as financial transactions which have been traditionally executed by a bank can now be shared by several parties. As an example, an Account Information Service Provider (AISP) can access a customer's accounts to display all their account information in one place. The Payment Initiation Service Provider (PISP) can initiate transfers on behalf of the customer of course both institutions must have the express consent of the customer.[77] Under the relevant legislation the FI must comply with the request of an AISP. The real problem is twofold first there is no dedicated dispute resolution mechanism in place[78], although Mastercard is delivering a product called 'Resolve' to "provide clarity, consistency and transparency for all participants in the Open Banking ecosystem"[79]and secondly in the end insurance

[74] Finextra, November 22, 2019, 'Trust in Open Banking: Negotiating data liability between banks and TPPs' <https://www.finextra.com/newsarticle/34820/trust-in-open-banking-negotiating-data-liability-between-banks-and-tpps>.
[75] Bill Streeter, September 18, 2019, The Financial Brand, 'Open banking scares consumers, but they want what APIs can deliver' https://thefinancialbrand.com/88283/open-banking-consumers-api-digital-data-privacy/
[76] Finextra (n 74).
[77] See https://www.finextra.com/blogposting/16647/open-banking-aisps-and-pisps-explained
[78] Finextra (n 74).
[79] Finextra, February 2019 'Open Banking: AISPs and PISPs explained' <https://www.mastercard.us/content/mccom/en-us/issuers/products-and-solutions/open-banking-solutions/resolve.html>; also see Mastercard, July 2 2020, 'Leading Firms Collaborate with Mastercard to Further Secure Open Banking Ecosystem' <https://newsroom.mastercard.com/eu/press-releases/leading-firms-collaborate-with-mastercard-to-further-secure-open-banking-ecosystem/>.

cover is essential as not only data mismanagement but also cyber theft needs to be addressed. The question is whether smaller PISP and AISP have the ability to be adequately covered for losses. The problem however is that "the FI is not permitted, under the regulatory framework, to make access contingent upon testing the TPP's robustness [...] In the EU and UK third parties need to pass the scrutiny of the regulator before they are permitted to access payment account data."[80]

There is an argument that mortgage brokers will become even more important than they are now. First of all, they provide a checklist of required items in setting out a Loan Application. The point for the assertion is that the educated consumer – as it was before - do not trust Lenders and this will not be changing, even if information is shared in an open banking situation. In the end it the confidence in the system which will drive consumer choices.

In sum three challenges must be overcome before the open banking system will function properly. First the interface – the API – as between the banks, consumers and TTPs must be safe and user friendly. Secondly, consumer information must be secure from fraud and consumers need to be aware of this security and thirdly the dispute resolution mechanism must be able to not only resolve potential disputes but must also be able to manage inquiries from all parties involved in the ecosystem.

Finally, trust by banks in the system also needs to be considered. For example, one problem that is likely to arise is that the banks will be the initial point of call for consumers where something needs to be addressed in their open banking experience, be it a complaint or a query. How the banks will be able to address these matters, whilst still retaining consumer confidence and satisfaction (in the bank itself and also open banking) will be an interesting task and one that has yet to be addressed. One possible way to address this could be for the relevant regulatory body to issue guidance to consumers, which outlines their rights in the open banking ecosystem, in addition to who to go to when issues arise. This guidance could also serve to inform consumers of the increased protection that open banking provides to them. It is worth noting that in 2019 in the UK, the Treasury Select Committee raised the point that a mandatory data protection code of conduct should be issued for open banking.[81] So far this has not gained

[80] Finextra (n 74).
[81] Payments Compliance, July 01, 2019, 'Treasury Committee Chair Calls for Mandatory Open Banking Code of Conduct' <https://paymentscompliance.com/premium-content/insights_analysis/treasury-committee-chair-calls-mandatory-open-banking-code-conduct>.

any traction. Another example is whether TPPs will prove themselves up to the ongoing task of fraud detection and protection. Banks are "watching closely and are very sensitive to how their future partners protect their customers".[82] If TPPs are not up to the task it may result in irreparable harm to them and to any hopes of longevity in the open banking ecosystem.

Given the trust in banks by consumers, it is the banks that should be making a concerted effort to 'vouch' for open banking to its customers. This has yet to occur and maybe it will take regulators to 'request' this of them before it transpires because it appears that banks need to see it as another regulation to adhere to, at least in the short term.

VI Implementation Concerns – Threat Or Opportunities?

The main issue certainly in the EU (leaving aside the interoperability of PSD2 and GDPR) is the lack of a common definition of what a standard API looks like. The effect is obvious not for the bank as such but for third parties who want and need to work with banks. "If every bank implements its API using different syntax and different data formats then building a solution that can access multiple banks becomes a large task."[83] and is costly for TTPs. To put it differently:

> There are multiple implementations of these APIs, which creates a need for a layer of aggregators in between. For commercial Open Banking the industry should work together to establish standards for the APIs – if we don't agree on the standards together, there will be another layer of aggregation that will add unnecessary cost.[84]

The end-effect is that TTPs have difficulty connecting to multiple banks which is the purpose of open banking and hence it is a potential drag on the adoption of open banking. Banks also must guard against the threat that they will not be paid for sharing the data as the PSD2 forces them to give an advantage to competing big techs and the data is given away for free.[85]

[82] Alex Rolfe, January 30, 2020, Payments Cards & Mobile, 'How fraudsters are attacking Open Banking weaknesses' <https://www.paymentscardsandmobile.com/how-fraudsters-are-attacking-open-banking-weaknesses/>.
[83] Mercator (n 20) 14.
[84] Mastercard (n 22) 7.
[85] Berger (n 10) 12.

It does not take much imagination to realize that if APIs are not streamlined, the cost-benefit analysis will indicate that traditional banking is to be preferred over open banking. A fragmented approach to API design will also affect security issues as TTPs would need to adjust to different levels of security. In sum the real problem at this time is that the infrastructure of open banking has not been clearly nor uniformly defined under the PSD2 nor in the Australian legislation. The main issue is risk management, that is who is ultimately responsible to safeguard data under the relevant data protection legislation? The unfortunate reality is that, especially in the EU, the financial sector is still occupied with compliance issues and to a lesser degree with product innovation, which is required in order to run a successful open banking ecosystem. The problem is exacerbated by the fact that compliance and data protection issues vary across the transnational field hence a bank with international branches may not be able at this time to use the same API as the regulation is not uniform (if at all present in certain countries). As an example, the US unlike the EU has taken a marketed-led approach to open banking.[86]

A *Security Issues*

Arguably in the short term at least, payment solutions advocated by Visa and MasterCard will be the forerunner of open banking not only in the EU but also in parts of the world such as the US and Australia. The real possibility is that some third parties like MasterCard will be playing an important role in the gradual implementation of open banking because they have found solutions to "a sophisticated fraud monitoring service, which provides real-time verification of third-party registration status, combined with fraud monitoring and alerts, to reduce financial institutions' exposure to liability and fraud)."[87] This issue is most important for banks they handle very sensitive information and as a result cybersecurity is of particular importance to them. Open banking in essence introduces a risk namely fraud, which can have significant effect on them and their bottom line by losing customers and reputation. This is especially of concern to banks as they are ultimately responsible even if a TTPs system is

[86] Mastercard (n 22) 7.
[87] Ibid.

breached. "It highlights the need for banks to be able to monitor and share information on potentially fraudulent activity across TPPs."[88]

This problem of validation of TTPs is even more acute once transnational transactions are taking place. The issue is not regulation but the fact that validation must be able to be up to date as it must guarantee security as real-time transactions and payments are the norm. MasterCard notes that:

> We may have national regulators that update their registers only on an intermittent basis, but yet all banks in Europe must accept requests from TPPs regulated in other member states. This is a potential risk.[89]

B *Dispute Resolution System*

One of the gaps in the PSD2 is the provision for a clear and complete dispute resolution system. This does not mean that the banking sector is not familiar with resolution systems specifically with other parties such as merchants or the bank itself especially the rulebook around card transactions. [90] However once TTPs are involved it elevates the potential dispute into a third layer. As in most cases real money is involved, the risk to sustain financial losses is real. One of the issues is when several banks are involved and not all have moved to real time payments, hence inaccuracies can occur which can result in losses to the client. The question will be asked who is responsible the bank or the service provider? A further complication can arise when an international transaction has taken place such as a TPP arranges for payment and the goods do not arrive. The question is contractual; that is who has standing to initiate dispute resolution proceedings? In addition, disputes can also arise in relation to data breaches and again the issue is where does the fault occur and who is responsible. Simply put the bank customer will initially request recourse from the bank even if the bank is not responsible but the bank is in effect the party, the consumer or client is familiar with. Importantly whatever system will be put in place." inability to properly and rapidly solve these issues will bring reputational damage not just to banks, but TPPs too. In both cases, the result would be a negative impact on overall confidence levels in open banking-enabled services."[91]

[88] Ibid., 8.
[89] Ibid., 9.
[90] Ibid., 10.
[91] Ibid., 11.

C *Opportunities*

In the EU the PSD2 also offers opportunities which have not yet been realised. They can be summarised into five clear areas. First, clients can be given new value-added services such as the creation of new personal finance management tools and the access to tailor -made TTP services which will affect the loyalty of customers and hence increase the bottom line. The key enabler in this area is the construction of an effective open API.

The second opportunity is that banks can become TTPs themselves and leverage other players data which includes other banks. It will therefore facilitate increased cross selling which can among other things simplify background checks.[92].

Thirdly and following from the above, a bank becoming a TTP they can attract new customers by offering debt restructuring at better rates.[93]

The fourth point relates to the increased use of open APIs which will affect the internal modernisation of processes and finally the PSD2 and other regulations will encourage financial players to move into the platform model. An example “BancoPosta (part of Poste Italiane) uses APIs to give clients access to a range of other banks' products, while at the same time offering its own services to other institutions.”[94]

VII Conclusion

Open banking in sum “aims to increase the level of security for customers and allow standardized access to certain customer data. The PSD2 levels the playing field between banks and third-party providers and benefits consumers by providing new services, increasing choice and enhancing security.”[95] The emergence of open banking has resulted in the unbundling and re-bundling of financial services was initially - and still is to an extent - driven by legislative compliance. However, “many of Europe’s banks have since transformed their response to PSD2 from that of achieving basic compliance to long-term strategic thinking about how to take advantage of the opportunities it presents.”[96] Banks

[92] Berger (n 10) 14.
[93] Ibid.
[94] Ibid.
[95]Ibid., 5
[96] Mastercard (n 22) 14.

are facing competition across the sector and the competitors are likely to be other banks and importantly neobanks which have the advantage of having a modern IT infrastructure already based on API technology.

There is no doubt that open banking is here to stay, and banks need to consider how to innovate and hence be competitive. However, at the same time it must be noted that, especially for second tier banks, the notion of face-to-face branch driven services might still be the best option to maintain competitiveness and grow at the same time. It enhances - what many banks lack – client-centricity an area in which big techs such as Amazon, Google or Facebook have shown strength. In addition, banks specifically ought to keep in mind what Schmidt and Cohen - Google executives - noted in their book namely that "the online world is not truly bound by terrestrial laws … it's the world's largest ungoverned space."[97]

That said it is the consumer who ultimately will judge the system and hence the choice of bank. The open banking system will no doubt rely heavily on strong and easy to access APIs with an attachment of progressive and customer focused TTPs. Simply put the financial services value chain will be shaken and changed with strong distinction between different providers in the forefront. Hence each financial provider, especially banks, will need to carefully select partners in order to deliver excellent services and make sure that issues dealt with in the Royal Commission into Banking[98] in Australia will not eventuate again. Clients in these circumstances can simply and easier than before change their financial service provider. However, the ability for banks to do their preferred level of due diligence on prospective TPP partners in the EU is somewhat limited by the requirement for them to allow access to regulated TPPs. In other words, their hands are tied once a regulated TPP comes calling with the requisite consent, but maybe lacking in other areas such as the requisite skill and knowledge to do the job that they have taken on. Given that open banking currently only addresses account and payment initiation, banks will not be losing much sleep. It will take a further movement, be it regulatory or market driven, before they invest any more time or effort in it.

The most important lesson is that inaction, that is sitting on the fence, will arguably eventually result in the demise of any tech company, bank or support

[97] Eric Schmidt, Jared Cohen, The New Digital Age: Reshaping the Future of People, Nations and Business, at 1

[98] See Royal Commission into Misconduct in the Banking, Superannuation and Financial Services Industry <https://financialservices.royalcommission.gov.au/Pages/default.html>.

agencies. This is so as TTPs cannot be prevented from entering what has been traditional banking territory and more importantly a TTP can be a competitor bank. "Ultimately … the future is one of an Open Banking world. But we are equally convinced that only those players that proactively shape their own future will succeed in it. PSD2 laid the regulatory foundations of such a new world – now it is up to banks to realize it. Those who do not, risk falling by the wayside."[99] One thing is for certain, the BigTech firms such as Amazon, Apple, Google and Facebook will continue being proactive in their involvement in financial services, which perhaps ominously often start with payments.[100]

[99] Berger (n 10) 26.
[100] Jon Frost et al, 2019, Bank for International Settlements, 'BigTech and the changing structure of financial intermediation, 2 <https://www.bis.org/publ/work779.pdf>.